

From Cross-Facility Detection to Metric 3D Localisation: A Monocular Perception Pipeline for Robotic Electric Vehicle Battery Disassembly

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Abstract—Robotic disassembly of end-of-life electric vehicle (EV) battery packs depends on perception that works on packs a system has never seen. Published detectors are almost always trained and evaluated on a single facility or a single rig, so their reported accuracy does not indicate whether they will transfer. We quantify this gap directly. A detector achieving 0.818 mAP@50 on the single-facility benchmark it was trained on falls to 0.277 mAP@50 on a 225-image benchmark spanning ten independent pack sources – a 66% relative loss. We then assemble a multi-source training set (4,425 images, 13 sources, ten pack families) and compare a from-scratch YOLO11n detector against RF-DETR, a real-time detection transformer with a frozen self-supervised DINOv2 backbone. Under an identical evaluation protocol RF-DETR attains 0.502 mAP@50 versus 0.410 for YOLO11n. Per-class analysis shows the two are statistically indistinguishable on conventionally annotated packs (module 0.771 vs 0.774); the entire advantage arises from graceful degradation under annotation-convention shift, where YOLO11n collapses (module 0.774 $\rightarrow$ 0.547) and RF-DETR does not (0.771 $\rightarrow$ 0.680). We further identify annotation convention as a measurable, dominant failure mode: per-source module mAP@50 ranges from 0.995 to 0.043, with the low outlier explained entirely by a divergent labelling definition rather than any visual difficulty. Finally, we show that condition assessment can be performed without any damaged training examples: a good-only anomaly detector over frozen DINOv2 patch features attains 0.857 recall on damaged modules, matching a supervised classifier trained on scarce damaged data. Finally, because a detection alone cannot command a manipulator, we close the loop to metric 3D: a fiducial-referenced homography lifts 2D detections into robot-frame coordinates from a single RGB view, with no depth sensor and no segmentation mask. Over 30 randomised trials on a URDF model of a real pack, localisation succeeded in 96.7% of trials at a mean 3D error of 5.67 mm (median 4.51 mm), an 87.6% reduction over a single-view baseline. We release the annotation set (16,945 polygon masks) and all evaluation code.

Index Terms—robotic disassembly, electric vehicle batteries, object detection, domain generalisation, monocular 3D localisation, fiducial markers, anomaly detection, recycling automation

I. Introduction

The volume of end-of-life EV battery packs is rising sharply, and disassembly is the bottleneck in both second-life reuse and material recovery. Disassembly is still predominantly manual: slow, costly, and hazardous, since packs carry high voltage and a thermal-runaway risk. Robotic disassembly is therefore an active research area, with recent systems demonstrating automated unfastening, module extraction and sorting [1]–[3].

Every such system begins with perception. Before a manipulator can unbolt, grasp or lift, it must locate the battery modules and the busbars that connect them. The prevailing approach is a supervised object detector trained on images of packs collected at one facility or one experimental rig.

This introduces a problem that reported accuracy conceals. A recycling line does not receive one pack type. It receives whatever arrives: different manufacturers, different module geometries, different states of disassembly, different lighting. A detector validated only on the distribution it was trained on provides no evidence that it will function on the next pack through the door.

A second problem follows immediately. Even a perfect detector returns pixels, and a manipulator needs metres. Bridging that gap is normally delegated to an RGB-D sensor, which adds cost, calibration burden and a failure mode of its own: the specular, dark, and highly reflective surfaces inside a battery pack are exactly the conditions under which consumer depth sensors degrade. We show the bridge can instead be built from a single RGB camera and planar fiducials placed once in the workcell.

We investigate both problems and make six contributions:

- 1) We quantify the cross-facility generalisation gap for EV battery module detection: 0.818 $\rightarrow$ 0.277 mAP@50, a 66% relative loss (Section V-A).

- 2) We show that a frozen self-supervised backbone (DINOv2, via RF-DETR) improves cross-facility detection over a from-scratch detector, 0.502 vs 0.410 mAP@50, and we isolate where the gain comes from: robustness to annotation-convention shift, not higher accuracy on conventional packs (Section V-B).
- 3) We identify and quantify annotation convention as a dominant, measurable failure mode, with per-source module mAP@50 spanning 0.995 to 0.043 (Section V-C).
- 4) We demonstrate condition assessment without damaged training data: a good-only anomaly detector reaches 0.857 damaged-module recall using zero damaged examples (Section V-D).
- 5) We close the perception-to-manipulation gap with a monocular, fiducial-referenced 2D→3D transform requiring no depth sensor, achieving 5.67 mm mean 3D error over 30 randomised trials on a URDF model of a real pack, and we show that multi-marker joint estimation reduces error by 75.6% over a single marker (Section V-E).
- 6) We report negative results that constrain the design space, including the failure of open-vocabulary detection (0.004 mAP@50) and of few-shot diffusion-based damage synthesis (Section V-G).

II. Related Work

A. Robotic EV battery disassembly

Zang et al. [1] survey the field and note that disassembly remains largely manual. Tan et al. [2] present a collaborative robotic system for unfastening and collecting screws that fuses vision with force/torque sensing under position and force control, reporting over 90% unscrewing efficiency. RAPID [3] integrates open-vocabulary detection with RGB-D perception on a gantry manipulator, reporting 0.9757 mAP@50 for component detection.

RAPID’s manipulation results are instructive for scoping any perception claim: with that detection accuracy, one-shot vision-driven grasping succeeded 57% of the time, rising to 83% with visual servoing and 97% with taught-in poses. Detection accuracy is necessary but far from sufficient for manipulation success. We therefore restrict our claims to perception.

B. Perception for battery components

Most published work targets fasteners. Screw detection has been addressed with Faster R-CNN and rotation edge similarity [4], with YOLOX-family models [5], and in a recent YOLOv11 versus Faster R-CNN benchmark [6]. Module- and busbar-level perception is comparatively under-explored; the closest work applies instance segmentation with point-cloud registration to grasp busbars from a single module type [7].

A recurring theme is data scarcity. Every study above collected and hand-annotated its own images – 1,719 images in [5] – and none released them. [6] resorts to

TABLE I
Principal data sources (13 total, ten pack families)

Source	Imgs	Pack family
ev-battery-comp-detect.	1,045	multi-manufacturer
ev-battery-components	680	Audi/BMW/Chevy
last_exp4	483	multi-manufacturer
ev-battery-pack	435	single lab rig
battery_comp	280	Nissan Leaf
ue_d1_defect	237	module close-ups
final_mobilenet	216	multi-manufacturer
automated-disassembly	117	disassembly cell
bmw_i3, ue_rav4	207	BMW i3, RAV4
Others (3 sources)	188	mixed

laptop screws as a proxy domain and states explicitly that the field is hampered by a lack of public datasets. This motivates our release of annotations.

C. Recovering metric pose from images

Vision-guided disassembly systems typically obtain depth from an RGB-D camera [3] or from point-cloud registration against a reference model [7]. Both add hardware and calibration cost, and structured-light and time-of-flight sensors are unreliable on the specular metal and dark polymer surfaces that dominate battery internals. Planar fiducials such as ArUco [12] offer an alternative: a marker of known geometry fixes a metric reference frame from a single view, and has been used widely for camera pose estimation and hand-eye calibration. We adopt this route, and quantify what it costs in accuracy.

D. Domain generalisation and foundation models

Self-supervised vision backbones such as DINOv2 [8] learn features that transfer across domains without task-specific fine-tuning, and detection transformers built on them, such as RF-DETR [9], are reported to generalise better than from-scratch detectors in low-data regimes. Whether that advantage holds on the narrow, cluttered, specular imagery of battery internals is an open question that we test.

III. Dataset and Benchmark

A. Sources

We assembled 4,425 training images from 13 public sources spanning ten EV pack families, summarised in Table I. Two classes are annotated: module and busbar.

B. Cross-facility test set

Reported accuracy is only meaningful against a benchmark that reflects deployment. We therefore construct a cross-facility test set of 225 images (780 annotated instances) drawn from all sources, held out entirely from training. We verified with perceptual hashing (dHash, Hamming distance ≤ 6) that no near-duplicate spans the train/test boundary for the sources we report per-source results on.

Deduplication proved necessary rather than precautionary. One candidate source, a 712-image public collection of 19 vehicle types, re-publishes images from another source in our set; 78 were byte-identical duplicates of our training images. Merging without this check would have leaked training data into the test set.

C. Annotation formats

Of 19,522 annotations, 16,945 are polygon instance masks (5–151 vertices) and 3,889 are axis-aligned boxes. We convert polygons to their enclosing boxes for detection evaluation. We note that an early version of our own pipeline silently discarded all polygon-format labels, training on 20% of the available annotations and producing an inflated, non-comparable score; we report this because the failure is easy to introduce and hard to notice.

IV. Method

A. Pipeline overview

The pipeline has three stages. Detection localises module and busbar instances in the image. Each detected module crop is passed to a condition-assessment stage that outputs a reuse grade: A (reusable), B (manual review), C (recycle), thresholded on the predicted probability of damage. Finally, a localisation stage lifts the 2D detection into metric robot-frame coordinates so that the result is directly actionable by a manipulator.

B. Monocular fiducial-referenced localisation

Let $\mathbf{K}$ be the intrinsic matrix of a calibrated monocular camera. We place N ArUco markers [12] of known side length at known positions on the workcell plane. Detecting their corners gives $4N$ correspondences between image points $\mathbf{u}_i$ and world points $\mathbf{X}_i$, from which we solve for the camera pose $(\mathbf{R}, \mathbf{t})$ by minimising reprojection error,

$$(\mathbf{R}^*, \mathbf{t}^*) = \arg \min_{\mathbf{R}, \mathbf{t}} \sum_{i=1}^{4N} \|\mathbf{u}_i - \pi(\mathbf{K}, \mathbf{R}, \mathbf{t}, \mathbf{X}_i)\|^2, \quad (1)$$

where $\pi(\cdot)$ is the perspective projection. Using all markers jointly rather than one at a time over-constrains the fit and averages out per-marker corner noise.

Because packs rest on a support of known height, the detected module centroid $\mathbf{u}_b$ back-projects to a unique 3D point: the camera ray through $\mathbf{u}_b$ is intersected with the plane $Z = h$,

$$\mathbf{X}_b = \mathbf{o} + \lambda \mathbf{d}, \quad \lambda = \frac{h - o_z}{d_z}, \quad \mathbf{d} = \mathbf{R}^{*\top} \mathbf{K}^{-1} \tilde{\mathbf{u}}_b, \quad (2)$$

with $\mathbf{o} = -\mathbf{R}^{*\top} \mathbf{t}^*$ the camera centre. The result is a metric grasp target in the robot frame obtained from one RGB image. No depth buffer, segmentation mask or CAD registration is used, and markers are never attached to the battery itself: the object is found by the RGB detector, and the markers only fix the reference frame.

TABLE II
Cross-facility generalisation gap

Model	In-domain mAP@50	Cross-facility mAP@50
Single-facility specialist	0.818	0.277
Multi-source YOLO11n	0.195	0.410
Multi-source RF-DETR	0.541	0.502

C. Detectors compared

Specialist. YOLOv8n trained on the single-facility source only. This represents the conventional approach and our own prior work.

Multi-source YOLO11n. YOLO11n (2.6M parameters) trained from COCO initialisation on all 13 sources, 150 epochs, 640 px, SGD.

RF-DETR. RF-DETR-Nano [9], a real-time detection transformer with a frozen DINOv2 [8] backbone, fine-tuned on the same multi-source data at 384 px for 60 epochs.

D. Evaluation protocol

Confidence scores are not comparable across architectures, so all models are scored with a single implementation (the supervision mAP implementation) at a common low confidence threshold of 0.05, on identical images, with identical ground truth including polygon-derived boxes. We stress this because mixing metric implementations across architectures silently produced a spurious result in our own early experiments.

E. Condition assessment

We compare a supervised ResNet18 classifier against a good-only anomaly detector: a PatchCore-style memory bank built from frozen DINOv2 ViT-S/14 patch features of healthy modules only. At inference, the maximum patch distance to the memory bank scores abnormality. This requires no damaged training examples, which matters because damaged modules are rare in any collectable dataset.

V. Results

A. The cross-facility generalisation gap

Table II and Fig. 1 quantify the central problem. The specialist detector scores 0.818 mAP@50 on the single-facility benchmark it was trained and validated on. On the cross-facility benchmark it scores 0.277 – a 66% relative loss. Its in-domain number is not predictive of deployment behaviour.

The inverse also holds: the multi-source YOLO11n scores only 0.195 in-domain on the specialist’s benchmark. Neither number alone characterises a model. Both regimes must be reported.

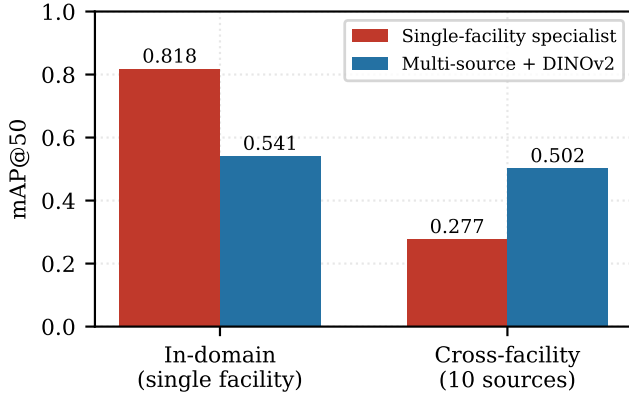


Fig. 1. A single-facility detector loses 66% of its accuracy when evaluated across ten pack sources. A multi-source detector with a frozen DINOv2 backbone is far more stable across the two regimes.

TABLE III
Detector comparison, cross-facility test (225 images)

Model	mAP @50	mAP @50–95	mod.	bus.
Specialist	0.277	–	0.231	–
YOLO11n (multi-src)	0.410	0.256	0.547	0.304
RF-DETR (DINOv2)	0.502	0.312	0.680	0.353

B. Detector comparison

Under the identical protocol of Section IV-C, RF-DETR reaches 0.502 mAP@50 against 0.410 for YOLO11n (Table III), and leads on the stricter mAP@50–95 (0.312 vs 0.256).

The aggregate figure is misleading on its own. Restricting evaluation to the seven sources that share a consistent annotation convention (177 images), the two models are effectively tied: overall 0.558 (RF-DETR) vs 0.544 (YOLO11n), module 0.771 vs 0.774, busbar 0.366 vs 0.344. The DINOv2 backbone confers no advantage on conventionally annotated packs.

The entire aggregate advantage therefore comes from behaviour under annotation-convention shift. Adding the outlier source drops YOLO11n’s module score from 0.774 to 0.547, but RF-DETR’s only from 0.771 to 0.680. The practical claim is narrow and specific: a frozen self-supervised backbone does not detect conventional modules better, but it degrades far more gracefully when the target definition shifts. For a recycling line receiving heterogeneous packs, graceful degradation is the property that matters.

C. Annotation convention as a failure mode

Per-source module mAP@50 for the multi-source detector is excellent on most packs – 0.995, 0.910, 0.873, 0.749 – and collapses to 0.043 on exactly one source.

Inspection shows the outlier is not visually harder. It annotates small internal sub-components in extreme macro

TABLE IV
Condition assessment (48-image held-out real test set)

Method	Damaged recall	Acc.	Damaged train ex.
Supervised ResNet18	0.857	0.792	80
DINOv2 linear probe	–	0.792	80
Good-only anomaly (ours)	0.857	–	0

views as “modules”, while the other sources annotate pack-level modules. The detector learned the majority convention and applies it consistently; the collapse is a disagreement about the target, not a failure of perception.

Two consequences follow. First, aggregate mAP over sources with inconsistent conventions is not a meaningful quantity, and per-source reporting is essential. Second, we attempted to repair the outlier automatically using an open-vocabulary grounding model, and it failed: on grayscale macro imagery the proposals were unusable. Convention conflicts are not cheaply fixable post hoc, which argues for fixing annotation guidelines before collection.

Busbar remains the weaker class for every model (0.30–0.37, Table III). We traced a probable cause: 61% of busbar instances in the test set come from a single source, and only four of eight sources annotate busbars at all. Busbar therefore suffers the same single-source annotation coverage problem that module detection suffers from convention conflict.

D. Condition assessment without damaged data

Damaged modules are rare in any collectable dataset; ours contains six distinct damaged packs. Table IV compares a supervised classifier against the good-only anomaly detector.

The good-only detector matches the supervised model’s 0.857 damaged-module recall while using no damaged training examples (AUROC 0.702). Because it models only the healthy distribution, it is not restricted to damage types seen during training. For a deployment where novel damage modes appear continuously and labelled damage is expensive, this is the more practical formulation.

Calibration analysis showed no damaged module was routed to Grade A (reusable), i.e. no false-safe assignments on the test set; expected calibration error improved from 0.284 to 0.246 under temperature scaling.

E. From detection to metric 3D localisation

A detection is only actionable if it can be expressed in the robot’s frame. We evaluate the localisation stage of Section IV-B in a PyBullet simulation with a calibrated monocular RGB camera, in three stages of increasing realism. Ground truth is read from the simulator for validation only; it never enters the estimator.

Single versus multiple markers. With one marker (ID 0, 0.12 m) over 30 trials, detection succeeded in 100% of

TABLE V

Monocular fiducial-referenced localisation (30 randomised trials each)

Configuration	Success rate	Mean 3D error (mm)	Median (mm)
Fixed-height baseline	—	45.65	—
Single marker	100%	7.35	6.97
Multi-marker (3)	100%	1.79	1.60
Full pack model (4)	96.7%	5.67	4.51

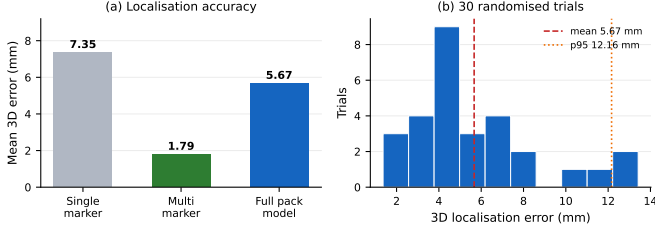


Fig. 2. Monocular localisation accuracy. (a) Joint multi-marker estimation cuts error by 75.6% over a single marker; the full pack model is harder than primitive geometry but stays under 6 mm. (b) Error distribution over 30 randomised trials on the pack model.

trials, marker localisation error was 4.05 mm mean, and the propagated error at the battery service point was 7.35 mm mean (median 6.97, max 15.93). Solving jointly over three markers on the same scenes reduced service-point error to 1.79 mm mean (median 1.60, max 4.91) with a joint reprojection RMSE of 0.47 px – a 75.6% relative reduction for no extra hardware. Over-constraining the pose fit is the single most effective change we made.

Full pack model. We then replaced the primitive geometry with a URDF model of a real 1S4P pack and randomised its placement over 30 trials ($x \in [0.375, 0.525]$ m, $y \in [-0.055, 0.055]$ m, yaw $\in [-25^\circ, 25^\circ]$). Results are given in Table V and Fig. 2. All four workspace markers were detected in 30/30 trials; the RGB detector found the pack in 29/30, giving an end-to-end localisation success rate of 96.7%. Mean 3D error was 5.67 mm (median 4.51, 95th percentile 12.16), decomposing into 3.25 mm in-plane and 4.07 mm along the optical axis, the expected anisotropy for a monocular method. Against a naive single-view baseline that assumes a fixed target height (45.65 mm mean error), this is an 87.6% reduction.

The residual error budget is informative: camera position error was 2.26 mm and workspace reprojection RMSE 0.354 px, so a substantial part of the 5.67 mm total is attributable to the reference frame rather than to the detector. Improving marker placement and calibration is therefore a more productive target than improving detection precision, which is not obvious a priori.

Placing this alongside the manipulation literature: RAPID reports 0.9757 mAP@50 yet 57% one-shot grasp success [3], indicating that the loss occurs between perception and actuation. A 5.67 mm target with a 12.16 mm

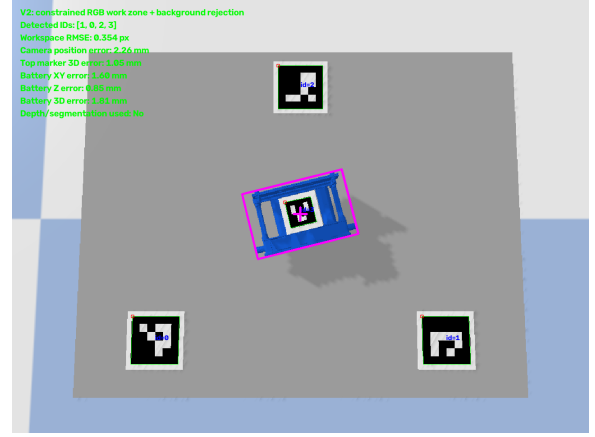


Fig. 3. Localisation scene. Workspace markers (IDs 0–2) fix the metric reference frame; the pack is found by the RGB detector (magenta) and never carries a marker used as an object label. Workspace reprojection RMSE 0.354 px.

95th percentile is within the closing tolerance of a compliant gripper on a module of this scale, suggesting the monocular route is viable for coarse positioning with final alignment delegated to force feedback or visual servoing.

F. Deployment

INT8 quantisation of the detector to ONNX reduced latency by 20% and model size by 46% (44.8 ms, 3.4 MB on CPU) for a loss of 0.007 mAP@50, which is compatible with edge deployment on a disassembly cell. The localisation stage adds only marker detection and a PnP solve, both negligible against detector inference.

G. Negative results

We report failures that constrain the design space:

- Open-vocabulary detection is unusable here. YOLO-World, prompted with “ev battery module” and “bus-bar”, scored 0.004 mAP@50 on the cross-facility test. Battery internals are too far out of distribution for current open-vocabulary detectors, notwithstanding their strong general-domain results.
- Naive multi-source merging degrades accuracy. Merging all sources without reconciling conventions reduced module mAP@50 to 0.331, below the single-source baseline. More data reduced accuracy.
- Few-shot diffusion damage synthesis failed. A LoRA-fine-tuned inpainting model trained on six real damaged packs produced 180 crops that all reproduced a memorised surface artefact rather than plausible damage. Below some diversity threshold, generative augmentation reproduces incidental features.
- Copy-paste and glare augmentation reduced detector mAP in our setting, despite being widely reported as beneficial elsewhere.

VI. Limitations

Our claims are about perception. We report metric localisation accuracy, not grasp success: no physical

manipulator was driven from these estimates, and as RAPID’s 57% one-shot grasp rate at 0.9757 mAP@50 shows [3], perception accuracy does not translate directly into manipulation success. Hand-eye calibration error, gripper compliance and contact dynamics are outside our scope.

The localisation results are obtained in simulation. Simulation isolates the geometry of the estimator from sensor noise, which is what we set out to measure, but it omits motion blur, rolling shutter, marker occlusion by the manipulator, and the specular highlights that make real battery internals difficult. Real-camera validation is required before these figures can be claimed for a deployed cell, and we expect degradation. The method also assumes packs rest on a support of known height; stacked or tilted packs violate this and would require depth or a second view.

The condition-assessment test set is small (48 images, 14 damaged), so those figures carry wide confidence intervals and should be read as indicative. Damaged-module labels in public data are also of uncertain provenance: on inspection, several crops labelled “damaged” showed no visible external damage, suggesting labels derived from pack-level provenance rather than per-module inspection.

Finally, our data is assembled from public sources rather than collected under controlled conditions, so pack-type coverage is uneven.

VII. Conclusion

Single-facility evaluation substantially overstates how well EV battery module detectors will perform in a recycling facility: a detector scoring 0.818 mAP@50 in-domain scored 0.277 across ten pack sources. Multi-source training with a frozen DINOv2 backbone raises cross-facility accuracy to 0.502 mAP@50, and we showed that this advantage is specifically robustness to annotation-convention shift rather than better detection of conventional modules. We identified annotation convention as a dominant and measurable failure mode, with per-source module mAP@50 ranging from 0.995 to 0.043. For condition assessment, a good-only anomaly detector matched supervised recall (0.857) using no damaged training examples.

We then closed the loop from pixels to metres without a depth sensor. A fiducial-referenced monocular transform localised a real pack model to 5.67 mm mean 3D error over 30 randomised trials at a 96.7% success rate, and solving jointly over several markers rather than one reduced error by 75.6%. Because most of the residual budget traces to the reference frame rather than to detection, calibration quality is the more productive engineering target.

Future work is real-camera validation of the localisation stage, integration with a manipulator under force feedback, and closing the label loop by combining worker annotations with electrical measurements taken on the line, which would supply the damage labels that public data lacks.

Data Availability

Annotations (16,945 polygon masks), evaluation code and trained detector weights are available at <https://github.com/Komil-jon/ev-battery-vision-pipeline>.

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